The Clockwise Prior: Convention Reversion in Language Models

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Abstract

Many quantities in science and engineering carry a sign fixed only by convention: a mesh current is positive if it circulates clockwise, a line integral is positive if the curve is traversed counter-clockwise, a cross product follows the right hand. Both choices are always physically valid, so an instruction to use the non-standard choice is legitimate and unambiguous. We study what language models do with such instructions across three physics and mathematics domains whose answers are verified to the sign by four independent solvers, eight instructable conventions, and sixteen model configurations spanning a 248× range of problem-solving competence. The central phenomenon, which we call *reversion*, is that models often solve the problem correctly and report the signs of the textbook-default convention rather than the instructed one. (The title's "prior" is a behavioral label for this default-seeking regularity, not an identified training-data mechanism.) Reversion is highly convention-specific. Posing the same problems under a counter-clockwise mesh-current instruction breaks 401 previously-solved problems across our panel while fixing 6, and the three most capable models revert on 70.6–77.9% of their correctly-computed variables; yet the same models comply almost perfectly with conventions whose reversal is itself part of the standard curriculum, and in every one of them the gap between the curriculum-absent reversal and every curriculum-taught reversal is at least 69.6 percentage points. Compliance is also load-sensitive and capability-gated: hardening problems within an obeyed domain raises each model's reversion (Gemma-4-31B from 5.3% to 35.8%) while a control instruction demanding the identical output change stays below 1%, and mid-scale models revert at 91–99% on conventions the capable trio obeys. Mitigation is conditional at best. A reasoning toggle that raises accuracy by up to 64 points moves reversion on most conventions by only a few points, restoring compliance only where the reversal is both taught and procedurally shallow; and one to three fully worked in-context demonstrations of the required convention change reversion by at most 13.6 points in one model, with no consistent improvement in the other two. Two predictions we pre-registered were refuted, and we report both. Grading is deterministic throughout, with no language-model judges.

1 Introduction

Ask a language model to analyze a simple circuit, but add one sentence: "mesh currents are defined counter-clockwise." Circuit textbooks almost universally draw mesh currents clockwise, but nothing in physics prefers one direction; the choice only fixes which currents are reported as positive. A model that follows the instruction produces the usual magnitudes with certain signs flipped. What we observe, in every capable model we test, is different: the model sets up correct equations, solves them correctly, recovers every magnitude, and then reports the signs that a *clockwise* analysis would produce. The requested format is followed and the magnitudes are right, so the substitution is not detectable from magnitude-only grading; only a sign-exact check against the instructed convention reveals it. (We grade the mandated answer line and do not audit the surrounding prose for caveats; a model that flagged its own substitution elsewhere in the response would still be scored as reverted.)

This paper names that behavior, measures it at scale, identifies three factors that predict it, and tests two mitigations that mostly fail. Throughout, we use one term for the central outcome, defined precisely in Section 2: a model *reverts* on a variable when its answer has the correct magnitude but carries the sign of the textbook-default convention rather than the instructed one. Reversion is a compliance failure that most evaluations cannot see, because they grade magnitudes, accept answers up to sign, or use judges that need not check the instructed frame.

Sign conventions are an unusually clean instrument for separating *competence* (can the model solve the problem?) from *compliance* (does the instruction win when it conflicts with training?). They are *arbitrary*: both choices are physically valid, so the instruction is never a trick. They are *binary*: on any single variable, compliance is a sign. And they are *exactly gradable*: a conventional solver computes the correct answer under either convention, so no rubric or judge model is involved. Because magnitude and sign are graded separately, one response yields independent measurements of both axes.

What we find. Our results are organized as four claims. (1) Reversion is *large, asymmetric, and convention-specific* (Section 3): flipping one convention instruction breaks 401 solved problems and fixes 6, while matched unusual-but-non-convention instructions are followed at 85–97%. (2) Three factors predict where reversion appears (Section 4): how the convention's *reversal* is represented in standard pedagogy, how hard the problem is, and how capable the model is. (3) Mitigation is conditional at best (Section 5): reasoning modes buy large competence gains and, on most conventions, little compliance; worked in-context examples buy little to nothing. (4) The framework is falsifiable and was falsified twice (Section 6): two predictions recorded in the project's version history before data collection came out opposite to our expectation, and both refutations are reported.

Why it matters. A sign is the difference between tension and compression, charging and discharging, stable and unstable. The failure mode we document is undetectable by magnitude-only grading, grows when problems get harder, and resists the first mitigation a practitioner would try. Evaluations that do not grade signs against the instructed frame will report such a model as excellent.

2 Definitions and instrument

2.1 Conventions, contracts, and cells

A *convention* is a sign or orientation choice fixed by custom rather than physics. A *convention contract* is the block of an instance's prompt that states which conventions the answer must use. Every instance states its contract explicitly, including the default one; models never have to guess. A *cell* is a version of the same underlying physics problem posed under a particular contract. The *default cell* states the textbook conventions. A *flip cell* changes exactly one convention and leaves everything else identical. Because the physics is shared, a model's behavior on the default cell and a flip cell of the same problem forms a matched pair, and causal claims about the flipped convention come from within-problem comparisons rather than across-problem averages.

A concrete example makes the design tangible. One circuits instance shows a five-element network and states, among other conventions, "mesh currents (i_1, i_2, i_3) are defined CLOCKWISE" (the default cell) or "…defined COUNTER-CLOCKWISE, i.e., the reverse of the usual clockwise choice" (the flip cell). The requested output is a fixed line of `name=value` pairs. Under the flip, the correct values of i_1, i_2, i_3 are the negations of their default-cell values; every other requested quantity is unchanged. A model that answers the flip cell with the default cell's signs has matched the textbook default rather than its instructions, and the grader detects this exactly.

2.2 Reversion, precisely

For every instance we compute two reference answers with conventional solvers: the *instructed answer* (correct under the stated contract) and the *default answer* (correct under the textbook contract). A variable is *diagnostic* for a cell when these two answers differ in sign; only diagnostic variables can reveal convention behavior, and all rates in this paper are computed over them. A model's value for a diagnostic variable is graded into exactly one of four categories: *compliant* (matches the instructed answer within a 5% relative tolerance), *reverted* (matches the *default* answer), *incoherent* (magnitude correct but matching neither reference), or *magnitude-wrong* (a competence error). The *reversion rate* of a model on a cell is the fraction of its magnitude-correct diagnostic variables that are reverted.

One property of this grading must be stated openly, because it bounds what we may claim. For the seven conventions whose transform is a pure sign flip, the instructed and default answers have equal magnitudes, so a magnitude-correct value has only two possible signs: the instructed one or the default one. On those conventions, "reverted" and "magnitude-correct with the wrong sign" are the same event by construction, and the incoherent category is reachable only under the study's single affine transform (the circuits reference-node shift). We therefore make no claim that sign errors are "directed" as a separate discovery; the informative quantities are the reversion *rates*, their contrast with matched controls, and the asymmetry of the paired design, all of which are falsifiable. (In the one cell where direction is testable, most magnitude-correct errors in fact land between the two frames; see Section 3.)

2.3 Three domains, eight conventions

Circuits: linear DC networks across seven topology families; requested variables are mesh currents, node voltages, and branch quantities. Its conventions are mesh direction (default clockwise), the passive sign convention, and the reference node. *Contour*: line integrals of polynomial vector fields around lattice polygons; requested variables are per-edge work integrals, total circulation, and flux. Its conventions are traversal orientation (default counter-clockwise), the sign of reported work, and the flux normal. *Vectors*: multi-body torque, angular-momentum, and magnetic-force problems in three dimensions; its conventions are cross-product handedness (default right-hand rule) and the Newton's-third-law reporting choice. Difficulty in every domain is a generator setting, and the contour domain ships an easy and a hardened tier drawn from widened generator settings (higher polynomial degree, more edges; the two tiers are distinct distributions, a caveat we return to in Section 4.2). The vectors domain is evaluated at its most hardened setting, which still leaves the capable trio near ceiling.

Two design choices matter later. First, the contour cells for clockwise orientation (`cw`) and for reporting work done *against* the field (`wrk`) demand the byte-identical output transform: negate every work value. They differ in which convention is named, and in one further respect that we do not control: the orientation convention participates in how each integral is set up, while the work-sign convention can be applied once at reporting time. The pair therefore matches the output demand exactly and the internal computational burden only approximately. Second, each domain poses every problem under two required solution procedures (mesh vs. nodal analysis; direct integration vs. Green's theorem; direct summation vs. the transfer theorem), which lets us check that reversion follows the reported frame rather than any particular procedure.

2.4 Verification and grading

Every accepted instance is solved by four independent routes before any model sees it: in circuits, two independently hand-derived formulations, an exact symbolic modified-nodal-analysis solver, and the NGSPICE simulator; in contour, symbolic integration of each parametrized edge, Green's-theorem double integration over an exact triangulation, an independently implemented rational-arithmetic engine, and numerical quadrature; in vectors, textbook component formulas, Levi-Civita summation, symbolic cross products, and floating-point evaluation, together with exact transfer-theorem identities. Three of the four routes use exact rational arithmetic and must agree to the bit; instances are rejected otherwise. The instructed answers for flip cells are additionally validated by *independent physical realization* rather than by re-applying the same transform: the reference-node cell by re-grounded re-simulation, the counter-clockwise and active-sign cells by re-derivation from raw observables, the left-handed cell by re-solving the entire problem with the Levi-Civita symbol negated, and the reaction cell by negating the force sources (125/125 problems pass in each domain). Grading is a deterministic parser over a mandated answer line; no language model participates in generation, verification, grading, or analysis.

2.5 Model panel

We evaluate sixteen configurations: GPT-5 at low reasoning effort, gpt-oss-120b, Gemma-4-31B, DeepSeek-V3, Llama-3.3-70B, Qwen2.5-72B, and the Qwen3 dense family (1.7B–32B), each Qwen3 size in both its reasoning and non-reasoning mode, which toggles behavior at fixed weights. Default-cell competence spans 0.4%–99.2% across configurations and domains, a 248× range. Decoding settings are uniform; serving stack and quantization are pinned and recorded per response. Because models were not served on a common inference stack, cross-model capability comparisons are descriptive; the fixed-weight Qwen3 reasoning toggle, which holds model, stack, and data constant, is the panel's strongest controlled contrast. We refer to GPT-5, gpt-oss-120b, and Gemma-4-31B as the *capable trio*; they are the panel's most capable members in every domain, though not uniformly near ceiling (GPT-5 at low effort solves 30.6% of circuit magnitudes, and the trio sits at 95–100% on vectors and 98–99% on easy contour).

3 Reversion is large, asymmetric, and convention-specific

Three facts establish the phenomenon. First, its *size*: under the counter-clockwise mesh instruction, the capable trio reverts on 70.6%, 74.2%, and 77.9% of their magnitude-correct diagnostic variables respectively (Table 1), and competence does not protect a model: Gemma-4-31B solves 85.2% of circuit magnitudes and still reverts on 77.9% of its diagnostic signs. Second, its *asymmetry*: in the within-problem paired design, flipping the mesh convention breaks 401 problems that the panel solved under the default and fixes 6; exact McNemar tests on these pairs are significant at $q_{\mathrm{BH}} \leq 0.006$ in every competent family. Third, its *specificity*: the same models comply at 85–97% with matched instructions that are equally unusual but touch no convention (Gemma-4-31B and DeepSeek-V3, the two models tested with these controls), and they obey several other convention flips almost perfectly (Section 4.1).

Two stability checks bound sampling noise. Re-running Gemma's counter-clockwise measurement five times on a resampled subset gives run-level rates between 82.0% and 94.1% (mean 86.2%); the primary full-panel figure is 77.9%. The run-to-run spread is a few points, so headline rates are stable, but item-level agreement across runs is only 39.6%: *which* problems revert varies from sample to sample even though *how many* revert does not. The rate, not any per-item behavior, is the reproducible object. In the hardened contour tier the corresponding five-fold resampling gives mean 34.4% (SD 6.4 points).

Finally, the one cell whose transform is affine rather than a pure flip deserves its own sentence, because it is the only place where the two reference frames differ in magnitude and error direction is genuinely testable: there, most magnitude-correct errors (74 of 83) match *neither* frame, consistent with partial application of the potential shift rather than wholesale frame substitution. We flag this as an open question about affine conventions rather than evidence either way about the sign-flip ones.

4 What predicts reversion

4.1 How the reversal is taught

The eight conventions differ sharply in how their *reversal* is represented in the material models learn from. We code each on a three-level ordinal scale by auditing standard textbooks (Appendix B lists the books and criteria; the audit was performed by one author after the vectors-domain result prompted it, a timeline we disclose because it makes the scale descriptive and hypothesis-generating rather than pre-registered). *Tier 0*: the reversal is taught as a routine operation (reversing a contour's orientation negates the integral; reaction forces; inward normals; work done against a field). *Tier 1*: the reversal appears as an error contrast or partial treatment (the left-hand rule as a warning; the active sign convention; non-ground references). *Tier 2*: we found no worked treatment of the reversal in any audited text (counter-clockwise mesh currents).

Table 1 reports per-model reversion for all eight conventions with problem-clustered bootstrap confidence intervals; we deliberately avoid pooling across models and avoid variable-level test statistics, since variables within a problem are correlated. The pattern is a strong ordering with honest wrinkles. In every one of the three models separately, the tier-2 convention's reversion exceeds that of every tier-0 convention by at least 69.6 percentage points (69.6 for GPT-5, larger for the other two). Tier-1 conventions are heterogeneous rather than intermediate (the reference node draws 1.2%–0.0%, handedness 0.6%–0.0%), and one tier-0 convention (orientation, at 5.3% for Gemma) sits above two tier-1 rates, so the scale orders the extremes cleanly but not the middle. With one convention at tier 2 and the coding performed post hoc, we present pedagogy as the strongest *orga-*

Table 1: Reversion for all eight conventions at each model's low-strain setting, by reversal-pedagogy tier (0 = reversal taught as an operation, 1 = taught as an error contrast or partial treatment, 2 = no worked treatment found). Confidence intervals are 95% bootstrap intervals clustering on the underlying problem.

		Reversion % [95% CI, clustered by problem]		
Convention (domain)	Tier	GPT-5	gpt-oss-120b	Gemma-4-31B
mesh direction (circuits)	2	70.6 [58.0,83.0]	74.2 [69.5,78.6]	77.9 [73.2,82.5]
passive sign (circuits)	1	0.0 [0.0,0.0]	11.6 [7.9,15.3]	3.4 [1.6,5.5]
reference node (circuits)	1	0.0 [0.0,0.0]	1.2 [0.0,3.9]	0.0 [0.0,0.0]
handedness (vectors)	1	0.6 [0.0,1.7]	0.0 [0.0,0.0]	0.0 [0.0,0.0]
orientation (contour)	0	1.0 [0.1,2.3]	1.0 [0.1,2.3]	5.3 [2.5,8.4]
reaction pair (vectors)	0	0.3 [0.0,0.7]	4.1 [2.3,6.1]	0.0 [0.0,0.0]
flux normal (contour)	0	0.4 [0.0,1.2]	0.0 [0.0,0.0]	0.0 [0.0,0.0]
work sign (contour)	0	0.0 [0.0,0.0]	0.1 [0.0,0.2]	0.0 [0.0,0.0]

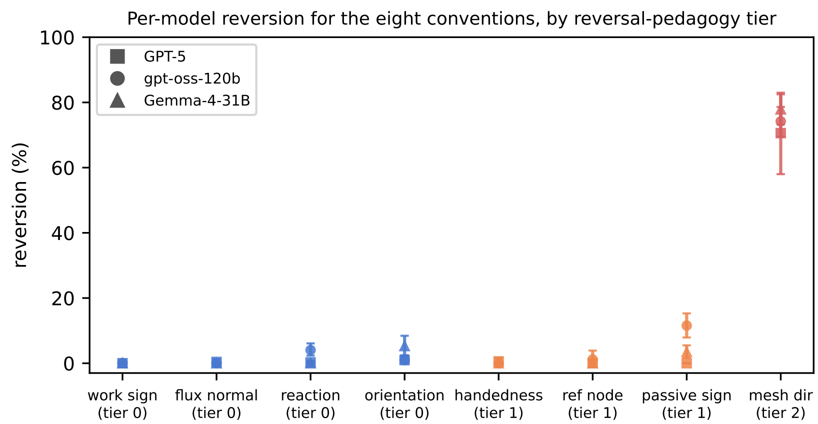

Figure 1: Per-model reversion for all eight conventions, grouped by reversal-pedagogy tier.

nizing variable we found, not as a demonstrated cause; the tier-2 pole's uniqueness is itself search-conditioned (Section 6).

4.2 How hard the problem is

Obedience on taught-reversal conventions has a load limit. We harden the contour problems (degree-3–4 fields, five-to-eight-edge polygons) and compare each model's reversion on the easy and hardened tiers: gpt-oss moves from 1.0% to 5.0% as its share of magnitude-wrong default-cell problems rises to 7%; GPT-5 from 1.0% to 13.7% at 24%; Gemma from 5.3% to 35.8% at 49% (five-fold resampling of this last figure: mean 34.4%, SD 6.4 points). The `wrk` control, which demands the identical negation of the same variables, stays at or below 0.3–0.4% for every model at every difficulty (Figure 2). The contrast rules out a generic inability to perform the required final negation under load; it does not by itself separate convention identity from differences in intermediate bookkeeping, since orientation participates in every integral while the work-sign flip applies once at reporting time.

Two caveats qualify the causal reading. The two tiers are distinct generated distributions (different seeds, polygon families, and rejection thresholds), so the comparison is between difficulty regimes rather than a controlled perturbation of identical items; and "strain" is the model's own error rate, a model-relative quantity, although the design knobs that produce it (degree, edge count) are set exogenously. Row availability also differs on the hardened tier (gpt-oss 197/250 default-cell rows, Gemma 183/250) because long generations truncate; we do not know the direction of the resulting bias and report availability rather than claiming conservativeness.

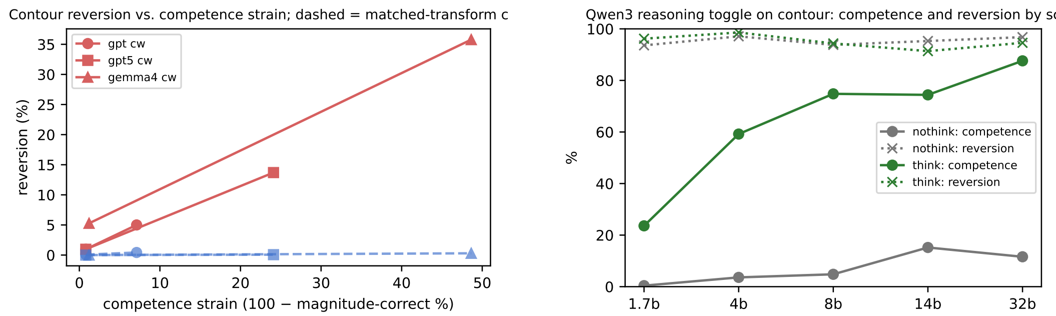

Figure 2: Left: contour reversion against competence strain; dashed lines are the transform-matched `wrk` control. Right: the Qwen3 reasoning toggle raises competence at every scale while orientation reversion stays above 90%.

Table 2: The Qwen3 reasoning toggle on contour: competence rises at every scale; orientation reversion does not follow.

	no-think		think	
Scale	Comp. %	Rev. %	Comp. %	Rev. %
1.7b	0	93.6	24	96.2
4b	4	97.1	59	98.6
8b	5	93.8	75	94.3
14b	15	95.3	74	91.4
32b	12	96.8	88	94.6

4.3 How capable the model is

Below the capable trio, compliance degrades even on tier-0 conventions: the Qwen3 family reverts on contour orientation at 91–99% at every scale in both modes (Table 2), and DeepSeek-V3, Llama-3.3-70B, and Qwen2.5-72B revert at 92–94%. Convention identity still matters at this floor (DeepSeek-V3: 92% on orientation versus 35% on the transform-matched work-sign cell), but capability ordering is not clean everywhere: on handedness, the 4B and 8B non-reasoning models have similar competence yet revert at 96.0% and 29.7% respectively, so scale alone does not determine convention behavior below the frontier.

5 What mitigation does and does not do

5.1 Reasoning modes: large competence gains, conditional compliance

On contour, toggling Qwen3's reasoning mode raises magnitude-competence from 0–15% to 24–88% across five scales while orientation reversion stays between 91% and 99% in every configuration (Table 2). In circuits we can check the same contrast at 32B, where the toggle raises competence substantially and leaves counter-clockwise reversion essentially unchanged; we did not run the smaller circuit think columns, so the circuits version of this claim rests on one scale. Deliberation, by itself, does not make the orientation or mesh instructions win.

The vectors domain shows the boundary of this negative result. On handedness, reasoning restores compliance: reversion falls from 96.0% to 2.9% at 4B and to 0.0% at 8B (Table 3, Appendix A). In the same responses, the reaction-reporting convention keeps slipping at 32.1%–83.4% across the three scales. A consistent reading of all three domains, which we offer as an interpretation of behavioral data rather than a mechanistic finding, is that deliberation can *retrieve* a reversal that is taught and procedurally shallow (a per-operation sign rule the chain of thought can state and apply), while it does not supply a reversal that was never taught, does not sustain one that must be maintained across a long derivation, and does not reliably hold a framing convention applied once at reporting time.

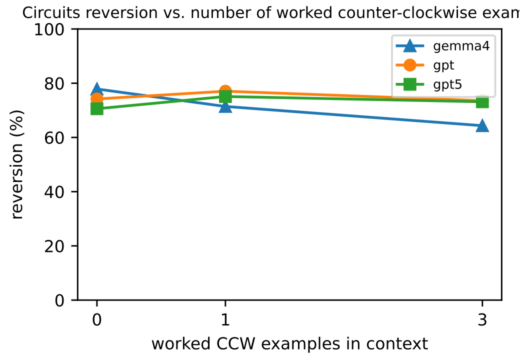

Figure 3: Circuits reversion against the number of in-context worked counter-clockwise examples.

5.2 Worked demonstrations: little to nothing

If reversion reflects a missing representation, supplying it in-context should help. We prepend one or three fully worked, solver-verified examples of counter-clockwise mesh analysis (held-out problems, different topologies) to the same circuit instances. Gemma improves by 13.6 points over the three-example curve (77.9%→71.4%→64.3%), roughly 4.5 points per example, a real but shallow effect at that exchange rate; gpt-oss (74.2%→73.6%) and GPT-5 (70.6%→75.1%→73.2%) show no consistent improvement (Figure 3). Because conditional denominators shift across arms, we also run a paired common-support analysis restricted to variables that are magnitude-correct in both arms of a comparison (Appendix C): on identical variables, Gemma's three-example arm moves 76.3%→62.4% (n=439; 113 fixed, 52 newly broken), gpt-oss is exactly flat (73.2%→73.2%; 26 fixed, 26 broken), and GPT-5's common support is too small for a paired read (n=25). GPT-5's baseline rests on 51 diagnostic variables against 526 and 190 for its one- and three-example arms, so its curve is read as flat-within-noise rather than as a precise trajectory. We tested one demonstration format on one convention; stronger interventions (explicit verification instructions, self-check scaffolds, fine-tuning) are untested and may behave differently.

6 Two pre-registered predictions, both refuted

Before collecting the corresponding data, we committed two predictions to the project's version history. **Prediction 1**: left-handed cross products would behave like counter-clockwise mesh currents, an untrained reversal, and the capable trio would revert heavily. *Refuted*: across the vectors panel (3,698 diagnostic variables at the most hardened setting, plus two easier calibration rounds), reversion on the left-hand rule is 0.0%–0.6% for the trio, with Gemma at 0.0% at 97.2% competence. The refutation prompted the pedagogy audit of Section 4.1: the left-hand outcome is taught, as a warning, in every audited text. We stress the epistemic status this gives the tier scale: it is a post hoc organization that the refutation motivated, not a confirmed prediction, and our failure to find a second tier-2 convention is a search-conditioned observation, not evidence of impossibility. **Prediction 2**: in-context demonstrations would collapse reversion if it stems from a missing representation. *Refuted* in two of three models; the third improved shallowly (Section 5). We report both refutations because a measurement framework that only confirms its authors' expectations is not measuring anything.

7 Related work

Instruction-following benchmarks grade compliance with formatting and content constraints (Zhou et al., 2023); counter-default instructions have been studied as inverse instructions (Stephan et al., 2025) and counterfactual task variants (Wu et al., 2024). Our contribution is machinery that separates compliance from competence within a single response using sign-exact oracles, with matched-transform and procedure controls. Conflicts between contextual evidence and parametric knowledge are studied in question answering (Longpre et al., 2021; Xie et al., 2024), and instruction hierarchies arbitrate conflicting directives (Wallace et al., 2024); our setting is the special case where the parametric side is a notational custom and the conflict is resolvable exactly. Evidence that procedural knowledge in pretraining shapes reasoning strategies (Ruis et al., 2025) is consistent with the curriculum-default account we hypothesize. Sycophancy work documents deference to user framing (Sharma et al., 2024); reversion persists with no user stance to defer to. Test-time-scaling studies measure capability gains from deliberation (Muennighoff et al., 2025; DeepSeek-AI, 2025); we show those gains transfer to compliance only narrowly. Context-aware decoding (Shi et al., 2024) and activation steering (Rimsky et al., 2024) modify behavior at inference time and are natural follow-ups we do not test. Known biases of model-based judging (Zheng et al., 2023) motivate our judge-free design, and our release protocol follows contamination-aware benchmark practice (Jacovi et al., 2023; Srivastava et al., 2022). The benchmark extends a published single-domain probe (Ravishankara, 2026a), follows deterministic evaluation practice from engineering-plot benchmarks (Ravishankara, 2026b), and uses canary-based contamination protection in the tradition of Srivastava et al. (2022); statistics use exact McNemar tests (McNemar, 1947) and clustered bootstrap intervals.

8 Discussion

Implications for evaluation. A benchmark that grades magnitudes, accepts answers modulo sign, or uses a judge that does not explicitly check the instructed frame will award full credit to every reverted answer in this study. Sign-exact grading against the instructed frame is cheap where the domain permits it and measures a dimension of controllability that accuracy does not.

Implications for deployment. The failure mode is invisible to magnitude-only grading, points toward the textbook default, worsens with difficulty, and is robust to the prompting mitigation we tested. A deterministic frame-check layer (recomputing the requested convention's signs from the model's own magnitudes) is a natural guard; we propose it as an engineering direction and note we have not evaluated it, and that affine conventions and coupled derived quantities would need care.

Limitations. The tier-2 pole is one convention, coded post hoc by one author from a finite textbook audit without blinded or independent coding; we present the scale as organizing, not causal. The interpretation of reversion as a *training* prior is an inference from behavior against textbook defaults; we never observe training corpora, and "curriculum-default convention" is the measured object. Strain is model-relative; the two contour tiers are distinct distributions; hardened-tier truncation censors rows in an unknown direction (availability is reported). Most columns are single-sample, with five-fold retests at two headline cells only; item-level behavior is unstable across resamples even where rates are stable. The 5% relative grading tolerance (with an absolute floor near zero) is not tuned: re-grading the headline cells at 1%, 2%, 5%, and 10% moves reversion rates by at most 2.5 points (Appendix C). The demonstration experiment covers one format on one convention. Our analysis is behavioral; nothing here localizes a mechanism inside the network.

Release. Generators, oracles, validators, grading, and analysis code are released publicly; every dataset regenerates byte-identically from a seed, so the public artifact is a one-command generator rather than a crawlable answer key. Frozen instances and model outputs are available under gated access with embedded canaries. We adopt this structure because our measurements are made against curriculum-default conventions: if these instances and their instructed answers entered future training corpora, the measurement itself, under our working hypothesis about its origin, would be altered rather than merely inflated.

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A Reasoning toggle on the vectors domain

B Textbook audit behind the pedagogy tiers

For each convention we examined the treatment of its *reversal* in standard textbooks: for circuits, Nilsson & Riedel; Hayt, Kemmerly & Durbin; Irwin & Nelms; Sadiku; and Kuphaldt's open text; for multivariable calculus, Stewart; Thomas; Marsden & Tromba; and the OpenStax Calculus series; for mechanics and electromagnetism, Halliday, Resnick & Walker; Serway & Jewett; Young & Freedman; Griffiths; and OpenStax University Physics. A convention's reversal is coded *tier 0* if at least one audited text states it as a rule or theorem and at least one provides exercises using it; *tier 1* if audited texts mention the reversal only as an error contrast or partial treatment without worked

Table 3: Qwen3 reasoning toggle on the vectors domain (hardened tier): competence % and reversion % per convention. Reasoning collapses the taught, shallow handedness reversal but not the reaction-reporting convention.

	no-think			think		
Scale	Comp.	lh rev.	rxn rev.	Comp.	lh rev.	rxn rev.
1.7b	16	89.2	93.0	37	35.9	83.4
4b	24	96.0	92.5	72	2.9	42.1
8b	23	29.7	22.1	79	0.0	32.1

examples; *tier 2* if no audited text provides any worked treatment of it. The audit was performed by a single author, after the vectors-domain refutation and before the drafting of Section 4.1; it is not blinded, has no second coder or agreement statistic, and should be read as a documented coding rather than a validated instrument.

C ROBUSTNESS ANALYSES

Unconditional outcomes. Table 4 reports, for every convention and capable-trio model, the shares of all four grading outcomes over *all* diagnostic variables (no conditioning on magnitude correctness).

Table 4: Unconditional outcome shares over all diagnostic variables.

Convention	Model	Compliant	Reverted	Incoherent	Mag.-wrong	n all diag
ccw	GPT-5	9.4	22.5	0.0	68.1	160
ccw	gpt-oss	17.4	50.2	0.0	32.4	642
ccw	Gemma	19.3	67.9	0.0	12.8	642
act	GPT-5	40.3	0.0	0.0	59.7	124
act	gpt-oss	47.4	6.2	0.0	46.4	500
act	Gemma	84.4	3.0	0.0	12.6	500
top	GPT-5	18.8	0.0	0.0	81.2	80
top	gpt-oss	48.5	0.6	3.4	47.6	328
top	Gemma	75.0	0.0	1.2	23.8	328
cw	GPT-5	98.6	1.0	0.0	0.4	1,350
cw	gpt-oss	98.6	1.0	0.0	0.4	1,350
cw	Gemma	86.8	4.8	0.0	8.4	1,325
wrk	GPT-5	99.4	0.0	0.0	0.6	1,350
wrk	gpt-oss	99.9	0.1	0.0	0.1	1,344
wrk	Gemma	99.6	0.0	0.0	0.4	1,344
inw	GPT-5	97.6	0.4	0.0	2.0	250
inw	gpt-oss	99.2	0.0	0.0	0.8	247
inw	Gemma	99.2	0.0	0.0	0.8	248
lh	GPT-5	98.5	0.6	0.0	1.0	1,248
lh	gpt-oss	99.1	0.0	0.0	0.9	1,242
lh	Gemma	98.6	0.0	0.0	1.4	1,248
rxn	GPT-5	95.3	0.2	0.0	4.5	1,248
rxn	gpt-oss	95.3	4.0	0.0	0.6	1,242
rxn	Gemma	99.0	0.0	0.0	1.0	1,248

Common-support pairing for the demonstration experiment. Table 5 restricts each comparison to variables magnitude-correct in both arms, removing denominator-composition effects.

Table 5: Paired common-support analysis of worked demonstrations (circuits, counter-clockwise cell).

Model	Arm	n common	Base %	Arm %	Fixed	Broken
GPT-5	1 example	43	74.4	65.1	7	3
GPT-5	3 examples	25	60.0	68.0	2	4
gpt-oss-120b	1 example	355	73.0	76.6	21	34
gpt-oss-120b	3 examples	299	73.2	73.2	26	26
Gemma-4-31B	1 example	485	75.9	71.3	99	77
Gemma-4-31B	3 examples	439	76.3	62.4	113	52

Grading-tolerance sweep. Table 6 re-grades the headline cells at four relative tolerances; rates move by at most 2.5 points.

Table 6: Reversion rate by grading tolerance.

Model	Cell	1%	2%	5%	10%
GPT-5	circuits ccw	70.6	70.6	70.6	70.6
GPT-5	contour hard cw	13.6	13.6	13.7	13.7
gpt-oss-120b	circuits ccw	73.8	74.2	74.2	74.0
gpt-oss-120b	contour hard cw	4.8	4.8	4.9	4.9
Gemma-4-31B	circuits ccw	77.8	77.9	77.9	77.7
Gemma-4-31B	contour hard cw	33.0	33.6	34.9	35.5

D REPRODUCIBILITY

All datasets are deterministic functions of a configuration file and a seed; regeneration commands, manifests with canary identifiers, transform-validation reports (125/125 per domain), the complete statistical freeze backing every number in this paper (including per-problem clustering used for all confidence intervals), and per-response provenance (model identifier, serving stack, quantization, sampling parameters) accompany the code release. No language model is involved at any point in generation, verification, grading, or analysis.